

TGA Assignment 03

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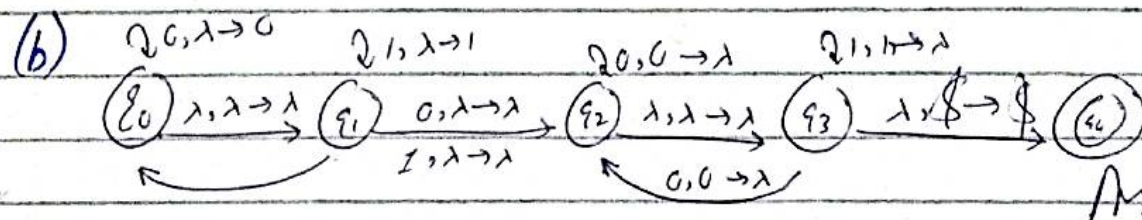
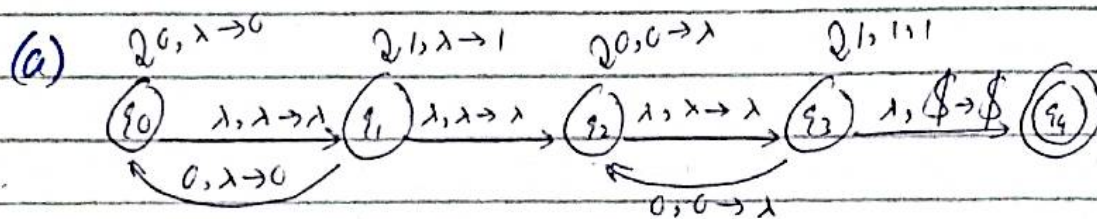
BLS-3H

Question #1:

(i) $S \rightarrow (S) \mid \Lambda$ Ans

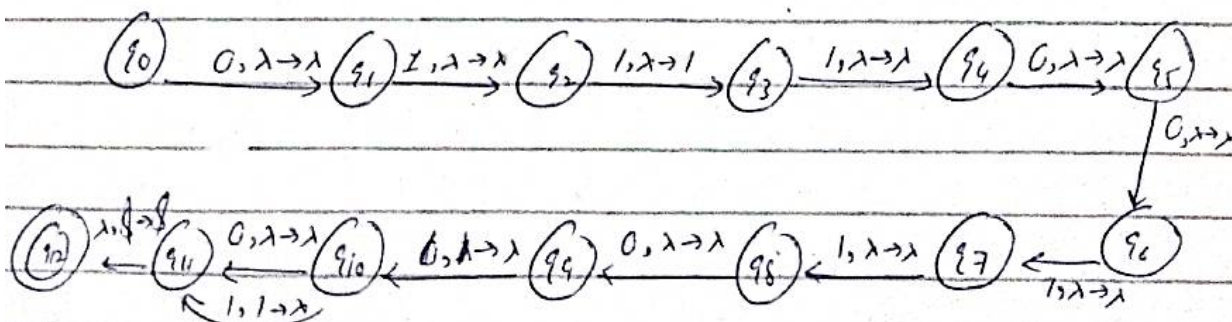
(ii) (a) $S \rightarrow OSO \mid ISI \mid \Lambda$

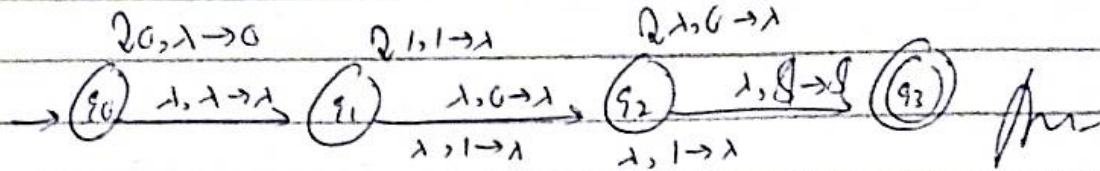
(b) $S \rightarrow OSO \mid ISI \mid \Lambda$



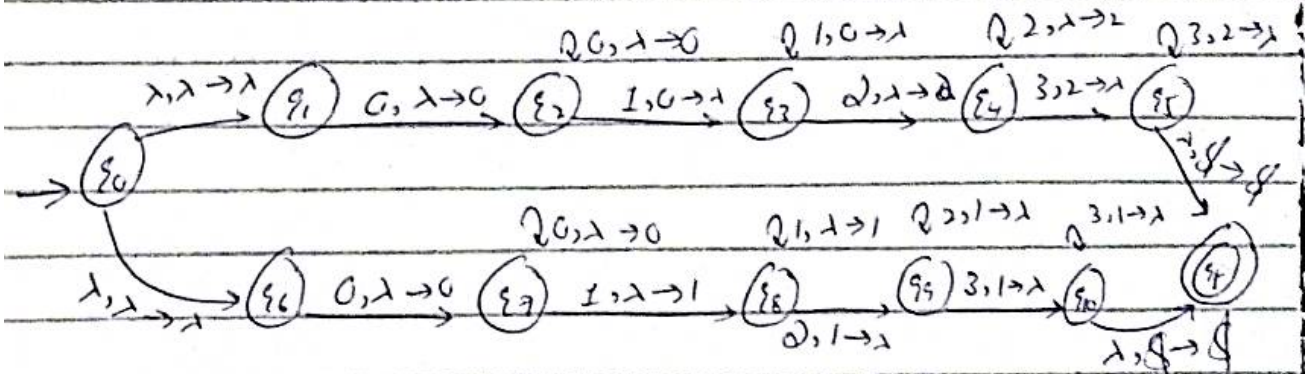
(iii) $S \rightarrow O \mid \Lambda$

$A \rightarrow 1100A10 \mid 110$



$$(iv) \begin{array}{c|c} S_1 \rightarrow OS_1 / OA & S_2 \rightarrow OS_2 I / IB \\ A \rightarrow OA / \lambda & B \rightarrow IB / \lambda \end{array}$$
$$S \rightarrow S_1 / S_2$$


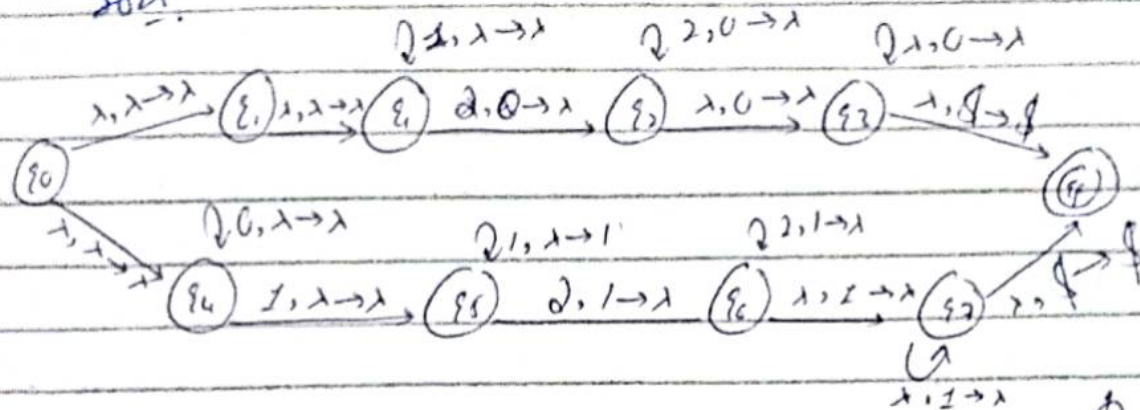
(v)
$$\begin{array}{l|l} S_1 \rightarrow AB & S_2 \rightarrow OS_23 / C \\ A \rightarrow CIA1 / OI & C \rightarrow IC2 / I2 \\ B \rightarrow OOB3 / OB & \end{array}$$



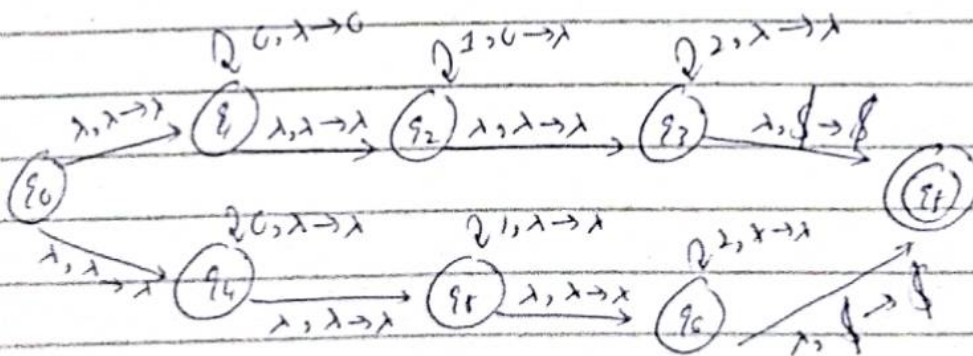
Ans

(vi) $S \rightarrow 0S_12 / 0S_11 / A \quad / \quad S \rightarrow BC$
 $A \rightarrow 1A1 \lambda \quad / \quad C \rightarrow 0B1 \lambda$
 $C \rightarrow 1C2 / 1C1 \lambda$
 $S \rightarrow S_1 / S_2$

Soln.



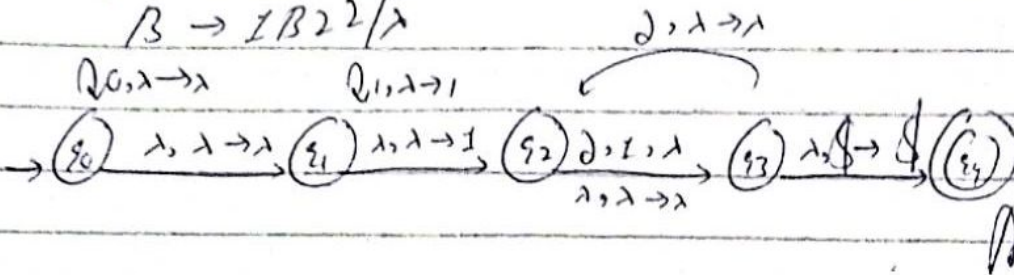
(vii) $S_1 \rightarrow AB \quad / \quad S_2 \rightarrow CD$
 $A \rightarrow 0A1 \lambda \quad / \quad C \rightarrow 0C1 \lambda$
 $B \rightarrow 2B1 \lambda \quad / \quad D \rightarrow 1D2 \lambda$
 $S \rightarrow S_1 / S_2$



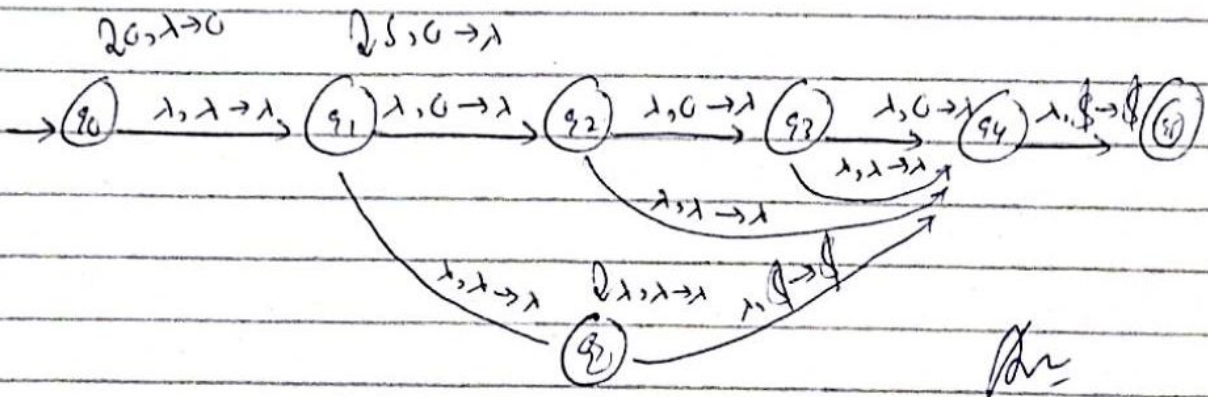
(viii) $S \rightarrow AB \mid \lambda$

$A \rightarrow 0A \mid \lambda$

$B \rightarrow 1B2 \mid \lambda$



(ix) $S \rightarrow 0S \mid 111S \mid \lambda$



(x) There is no cfy & no PDA that accepts that language. This language is not Context-free.

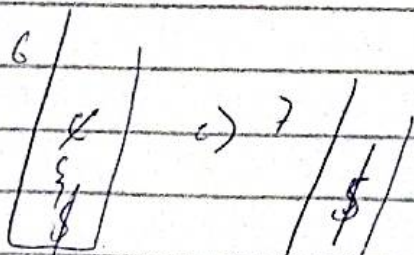
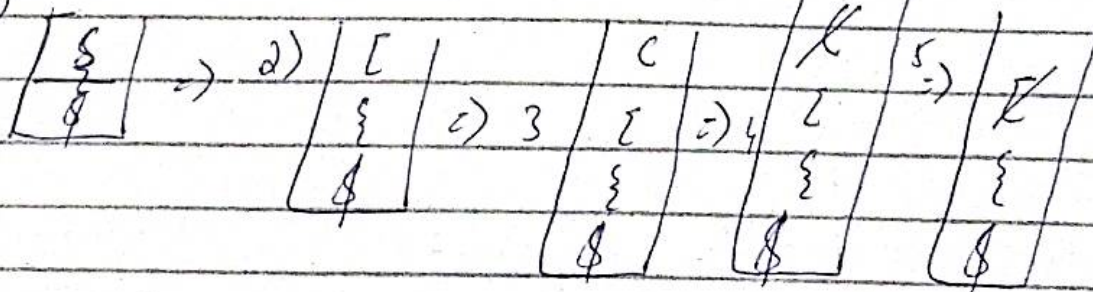
(c)

$z_0 \xrightarrow{ '(' } \text{push '('} \rightarrow z_0$
 $z_0 \xrightarrow{ '{' } \text{push '{' } \rightarrow z_0$
 $z_0 \xrightarrow{ '[' } \text{push '['} \rightarrow z_0$
 $z_0 \xrightarrow{ '\lambda' } \text{push nothing} \rightarrow z_1$
 $z_1 \xrightarrow{ ')' } \text{pop ')} \rightarrow z_1$
 $z_1 \xrightarrow{ '}' } \text{pop '}' \rightarrow z_1$
 $z_1 \xrightarrow{ ']' } \text{pop ']' } \rightarrow z_1$
 $z_1 \xrightarrow{ '\lambda' } \text{nothing} \rightarrow z_1$
 $z_1 \xrightarrow{ '\lambda' } \text{pop \& push \&} \rightarrow z_0 \text{ (final)}$

Ans

(d) Suppose in my input lab I have $\{ [()] \}$
 $[()] \}$ $(()] \}$ $)] \}$ $] \}$

1)



→ poped → $([\}$

(e) CFG:-

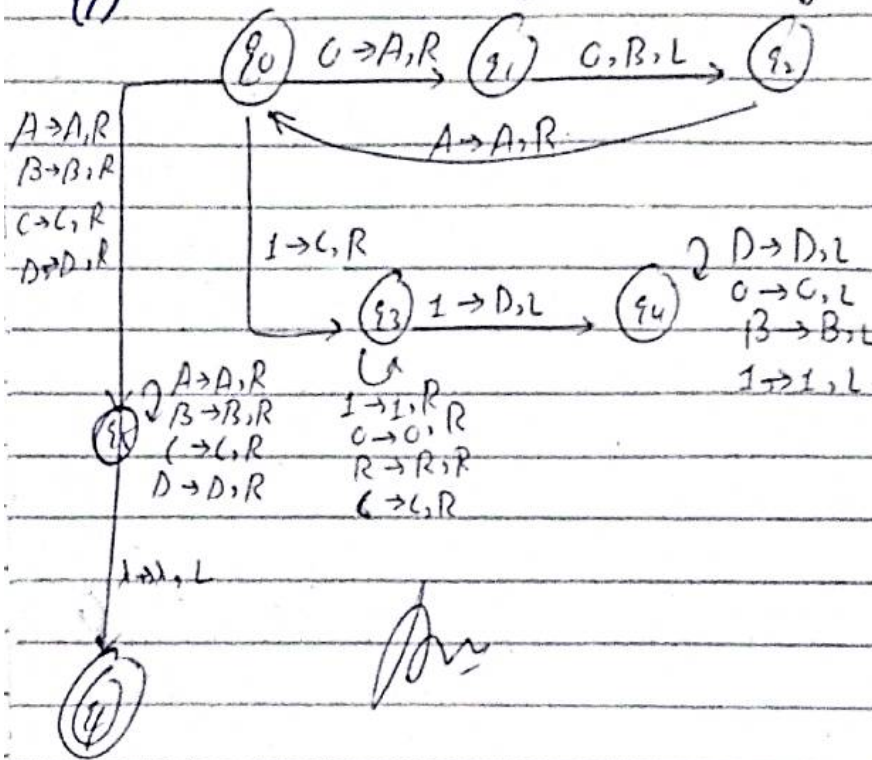
$S \rightarrow (s) \mid \{s\} \mid [s] \mid \lambda$ Ans.

Question #3:

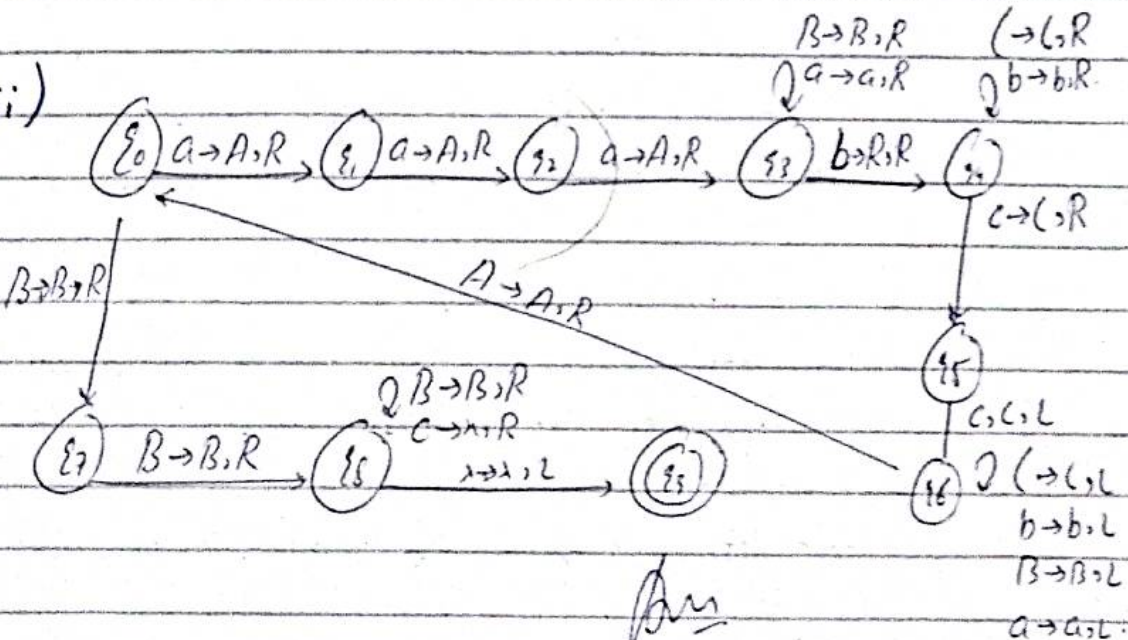
$1 \rightarrow 1, B$
 $C \rightarrow C, R$
 $B \rightarrow B, R$
 $A \rightarrow A, R$
 $C \rightarrow C, R$

$C \rightarrow C, L$
 $I \rightarrow C, L$
 $1, I \rightarrow L$
 $B \rightarrow B, L$

(i)



(ii)



(iii)

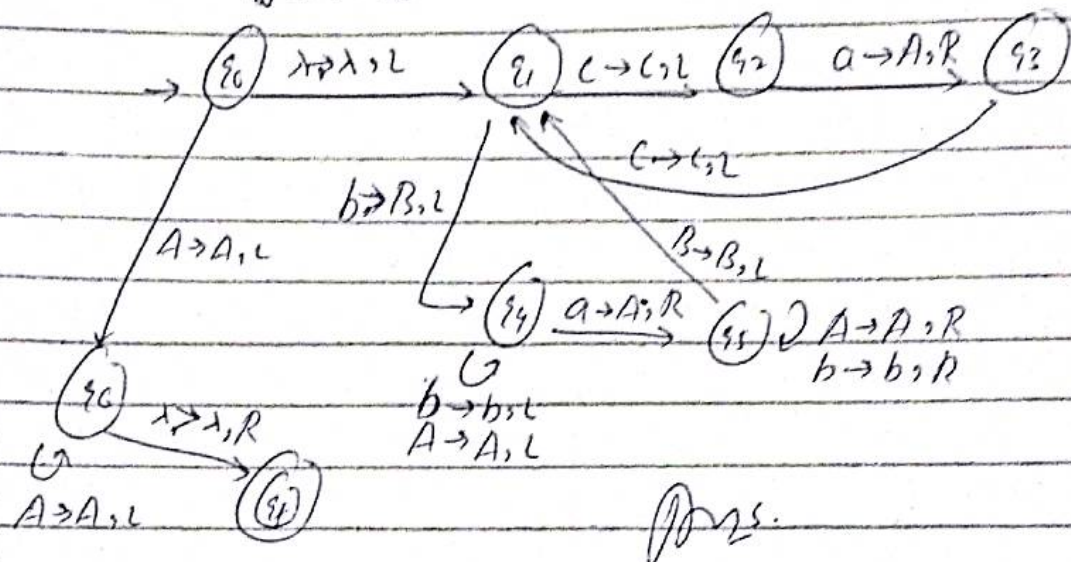
Q2.

$C \rightarrow C, R$
 $b \rightarrow b, R$
 $a \rightarrow a, R$

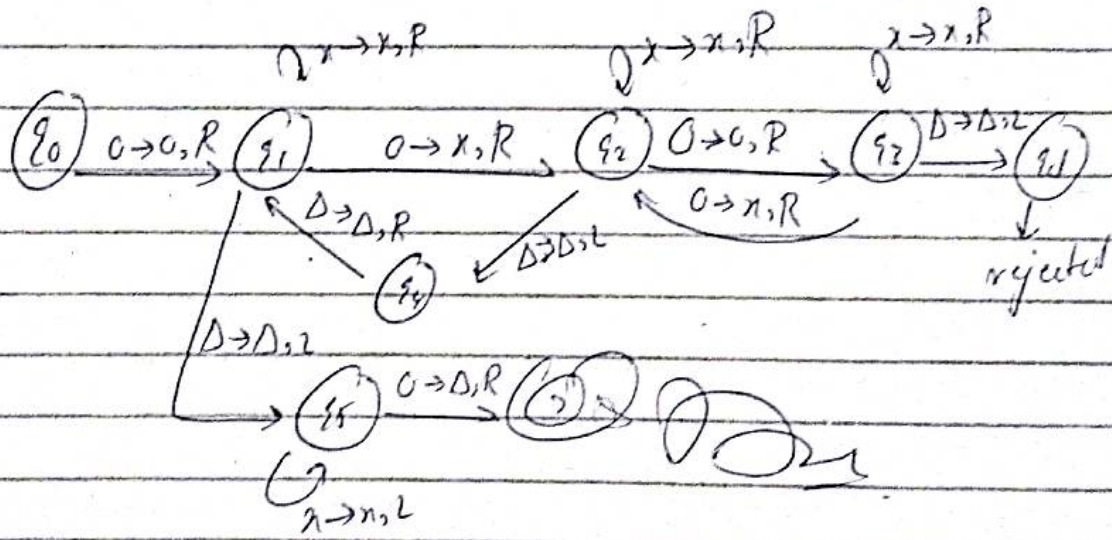
$A \rightarrow A, L$
 $B \rightarrow B, L$

$b \rightarrow b, L$
 $C \rightarrow C, L$

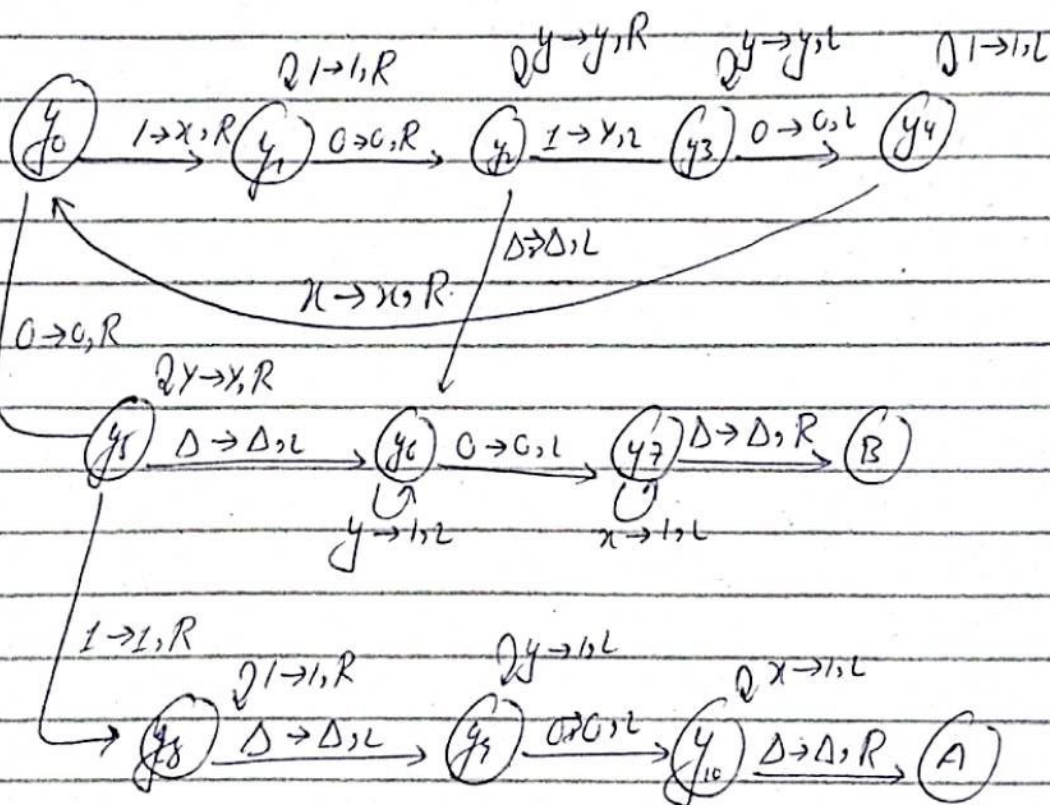
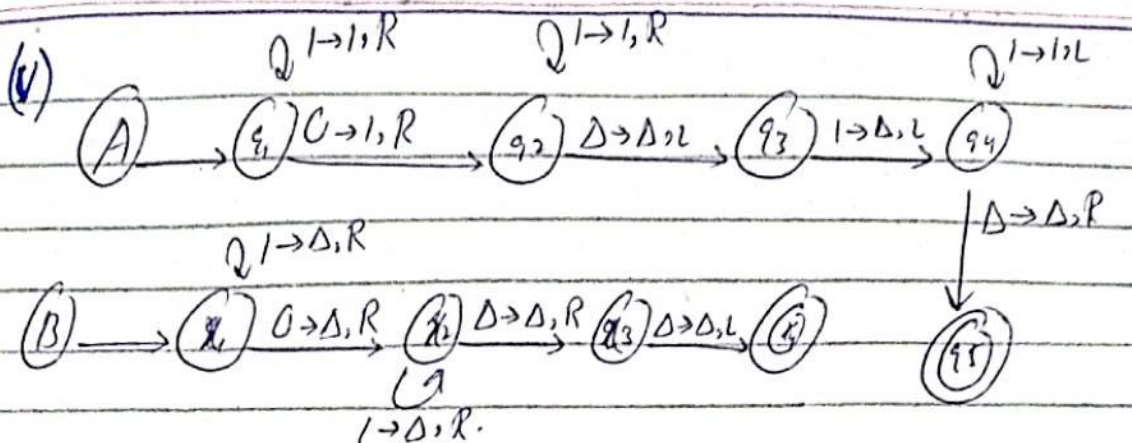
$A \rightarrow A, R$
 $C \rightarrow C, R$
 $B \rightarrow B, R$
 $b \rightarrow b, R$



(iv)



$\rightarrow A, R$
 $\rightarrow C, R$
 $\rightarrow B, R$
 $\rightarrow b, R$

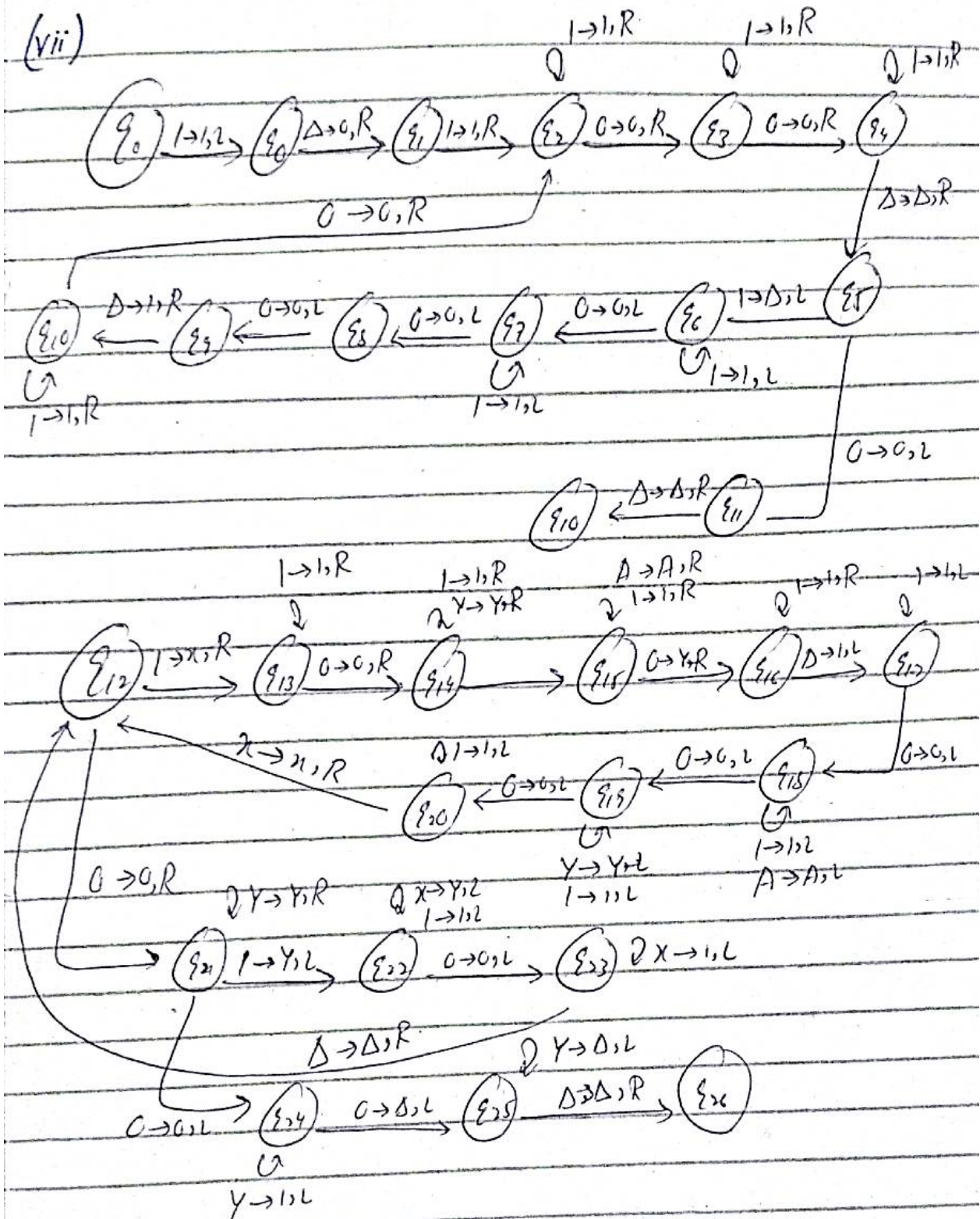


$\rightarrow \Delta, L$
 $\rightarrow \rightarrow$
 rejected

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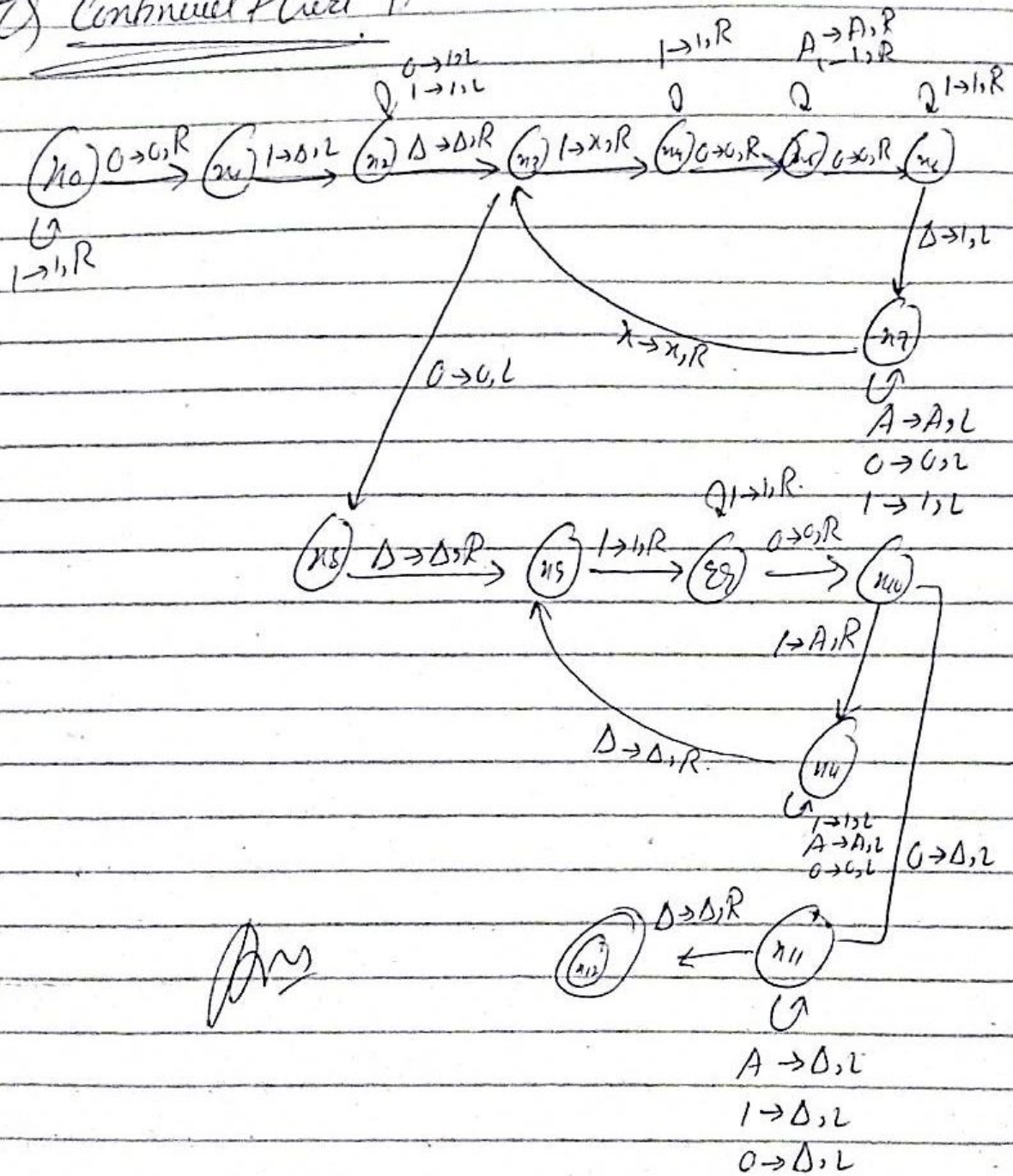


(vii)

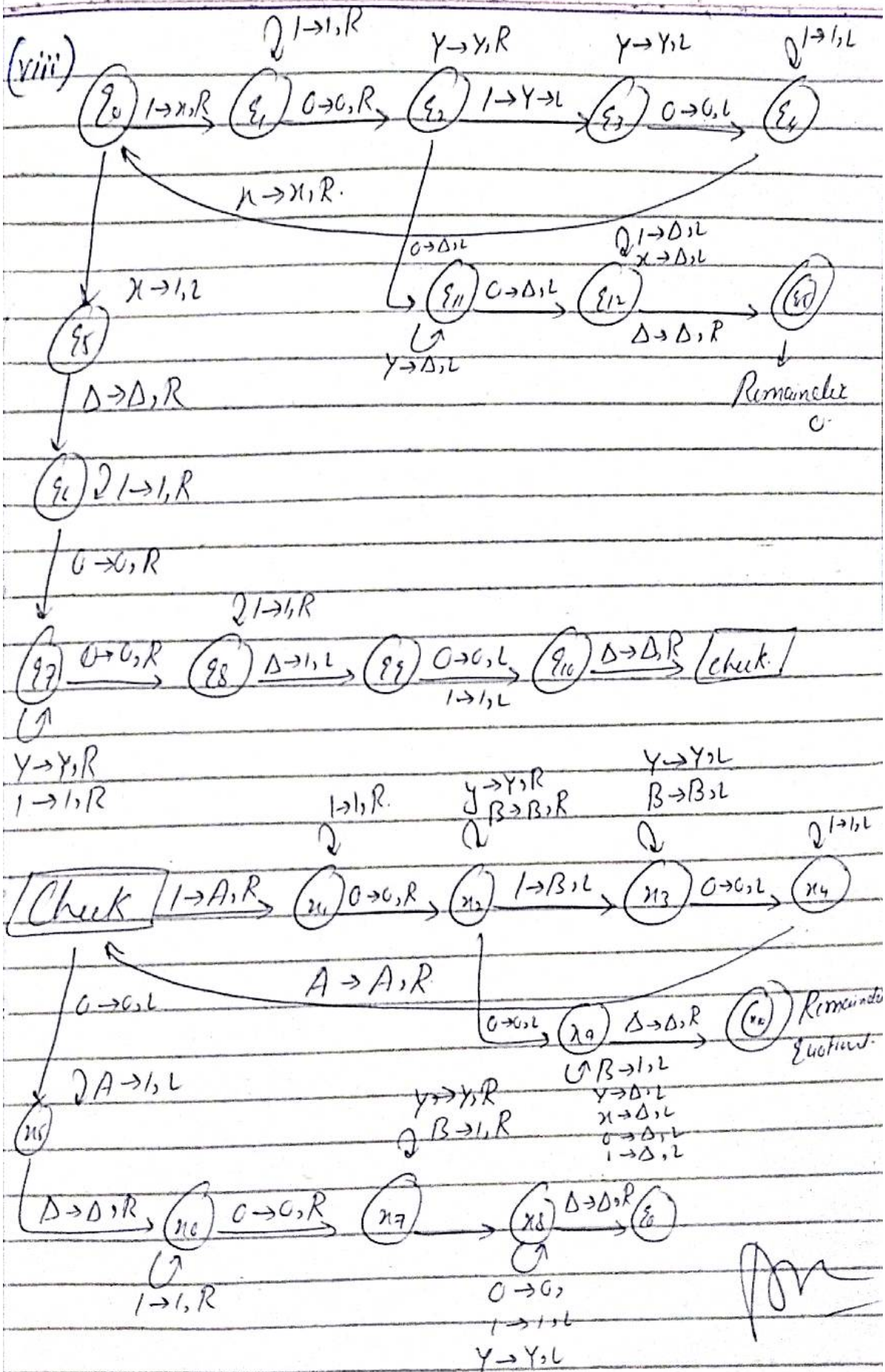


To be continued.

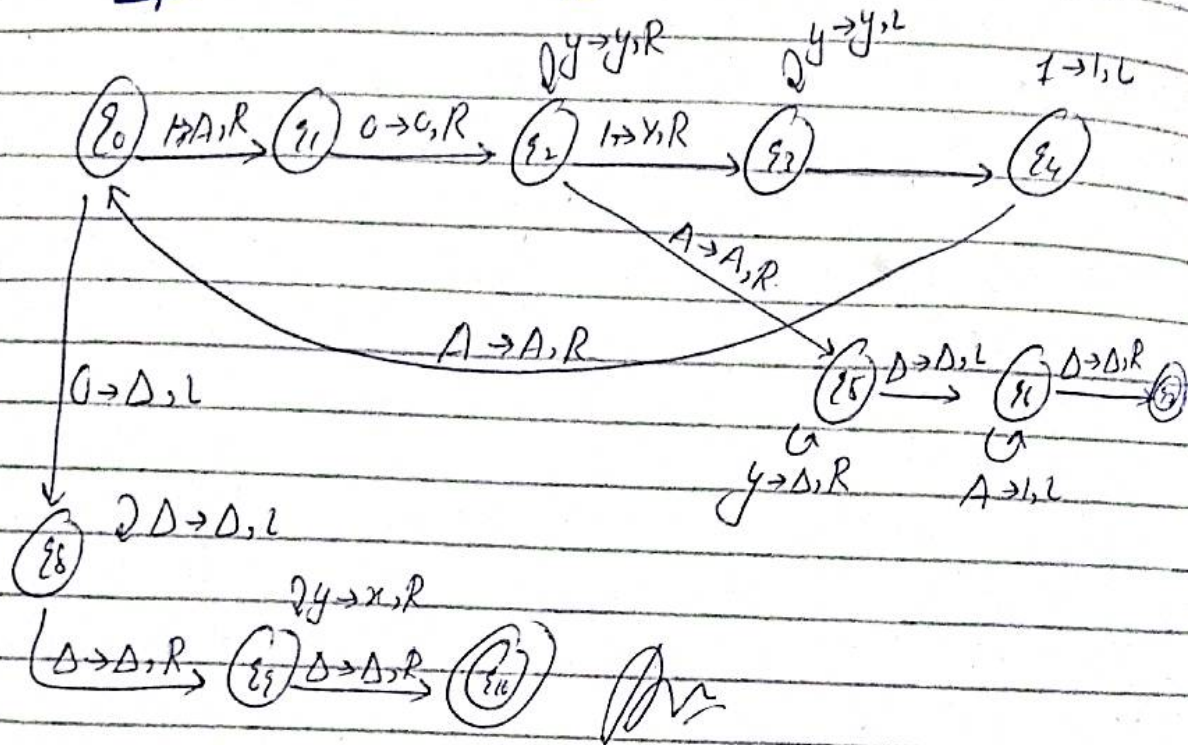
Continued Here 1



Am



(ix) Input:- 11110111 or 11101111



(x)

